

Quantum Kick

Developed by Rolands Frīdemanis

Game description

QUANTUM PUZZLE

Transform every alien into |0⟩.

- **Enemies** - aliens carry quantum state
- **Defense** - quantum gate traps on 4 lanes
- **Measurer** - measures every alien at the lane end
- **Goal** - |0⟩ is destroyed, anything else costs health



Current state

IMPLEMENTED

- **Qubit simulator** - states, gates, measurers
- **Five gate traps** - X, H, Z, S, T
- **Seven alien states** - |0⟩, |1⟩, |+⟩, |−⟩, |i⟩, |−i⟩ and T|+⟩
- **Waves**
- **Live forecast** - probabilities of *kills*
- **Score, health and high score**

CURRENT GAMEPLAY



1 **Build**
Preview info & build

2 **Wave**
Aliens walk on gates

3 **Measure**
Alien states are measured

4 **Repeat**
Go to next wave or lose

Quantum concepts

QUANTUM MECHANICS

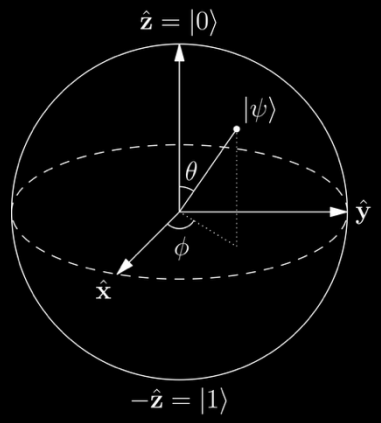
- **Qubits** - every alien is a quantum state
- **Superposition** - |+⟩ aliens make outcomes uncertain
- **Measurement** - the detector decides kill or leak
- **Interference** - the right gate turns 50/50 into 100%
- **Phase** - same odds, different fix: gate order matters

GATES

X	H	Z	S	T
Pauli-X	Hadamard	Pauli-Z	Phase S	Phase T

IMPLEMENTATION

- **Simulator** - built from scratch, complex amplitudes
- **Gates** - 2×2 matrices multiplied into the state
- **Measurement** - random draw weighted by |amplitude|²



Bloch sphere

State → (x, y, z) after every gate
Sets the alien's color and label

Yet to be implemented

PLANS FOR THE FINAL VERSION

- **Entanglement** - paired aliens across two lanes
- **CNOT gates**
- **Full scoring** (multipliers)
- **Game effects / UI improvements**

AI usage

Claude Code

Anthropic's AI coding agent

BUILD

Implementation via TypeScript one feature at a time.

TEST

Writing unit tests for the quantum simulator and game rules, and debugging failures.

PRESENT

Drafting and laying out the content of this presentation.

Every AI change was reviewed and play-tested.